



ENGINEERING / CTE

Introduction to Engineering Design Challenge

College - Introductory Engineering | Project Packet

Design a paper structure using at most 10 sheets, supporting 500 g for 10 seconds on a low tabletop. Record three trials, inspect failures, and justify one redesign. Use small classroom-safe weights under instructor supervision.

INSIDE THIS RESOURCE

Teacher guidance • 16 practice prompts • Reflection exit ticket • Complete answer guidance

Print the practice pages separately from the teacher key. Provide extra paper for extended work.



Teacher guide

Course objective - Define a design problem, distinguish criteria from constraints, generate and compare concepts, plan a fair test, and justify an evidence-based iteration.

Suggested time: 60-90 minutes

Learning focus: Design a paper structure using at most 10 sheets, supporting 500 g for 10 seconds on a low tabletop. Record three trials, inspect failures, and justify one redesign. Use small classroom-safe weights under instructor supervision.

Teaching sequence: Review prior instruction and model the first prompt with its answer. Assign one four-prompt page at a time. Ask for a check, a reason, or a specific example. Use the reflection ticket to plan the next lesson; it is not a standardized mastery score.

Model prompt: Define criterion.

Model answer: A desired feature or measure of success.

Support options

AIG: justify a second method or create a transfer example. 504: follow the learner's plan for time, breaks, format, and reduced copying. EC: model steps and accept aligned oral, drawn, or written responses. ESL: preview vocabulary, use bilingual references, and rehearse before writing.

Source and scope: Course-level learning objective; no accreditation, transfer, or universal equivalency claimed.



Practice 1–4

Show work or explain your response. Use additional paper when needed. Teachers may cut apart the four bordered prompts.

1. Define criterion.

2. Define constraint.

3. Given max 10 sheets, identify the constraint.

4. Given support 500 g, identify the criterion.



Practice 5–8

Show work or explain your response. Use additional paper when needed. Teachers may cut apart the four bordered prompts.

5. Why generate multiple concepts?

6. Name a fair-test variable to control.

7. What is a prototype?

8. Why record failure location?



Practice 9–12

Show work or explain your response. Use additional paper when needed. Teachers may cut apart the four bordered prompts.

9. Create a weighted decision matrix with three criteria.

10. What makes test evidence reliable?

11. Distinguish accuracy from precision.

12. Suggest one safety rule.



Practice 13–16

Show work or explain your response. Use additional paper when needed. Teachers may cut apart the four bordered prompts.

13. How can triangles strengthen structures?

14. State a testable claim about folded paper.

15. Use evidence to justify an iteration.

16. Reflect on tradeoffs.



Reflection exit ticket

Complete independently. Refer to your completed practice, but explain in your own words. This checks reflection and application, not a test score prediction.

1. State the focus of this resource in your own words.
2. Create one accurate example connected to the practice.
3. Explain how you checked that example.
4. Identify one possible error and show a correction.
5. Name one specific next practice step and explain why it will help.



Teacher answer guidance

1. A desired feature or measure of success.
2. A limit the design must obey.
3. Use no more than 10 sheets.
4. Support at least 500 g.
5. To compare alternatives before committing.
6. Same load placement, materials, span, or test time.
7. A testable representation of a design idea.
8. It reveals where and how to improve the design.
9. Answers vary; criteria, weights, scores, totals required.
10. Consistent procedure and repeat trials.
11. Accuracy is closeness to true value; precision is repeatability.
12. Keep loads low, protect hands/eyes, and clear the test area.
13. They resist shape change and distribute loads.
14. Answers vary; must identify design change and measurable outcome.
15. Cite test result, failure mode, and planned change.
16. Explain how improving one criterion may reduce another.

Reflection criteria

Items 1–4: 2 points for an accurate, specific response; 1 for a partly correct response; 0 for a missing or unrelated response. Accept justified alternatives. Item 5 is an unscored planning reflection. Feedback should identify a next teaching move, not label a learner.

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